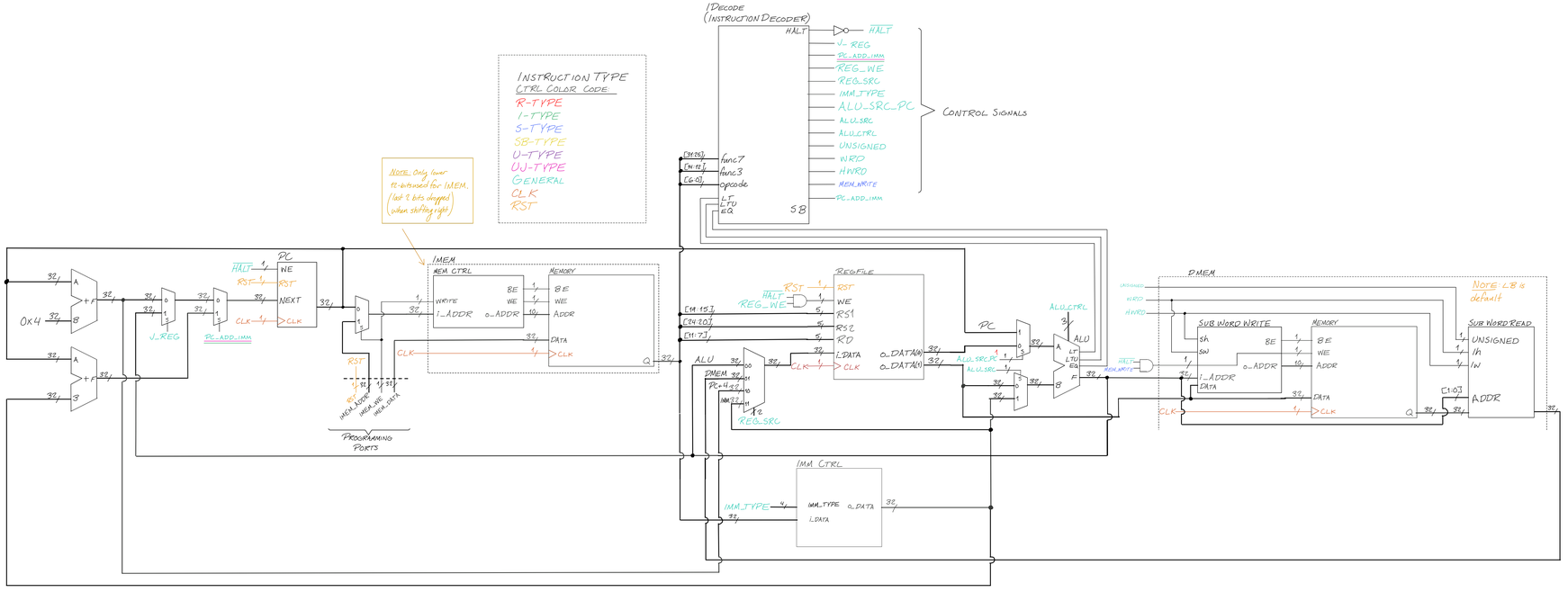

IDecode
(Instruction Decoder)
HALT
J_REG
PC_ADD_IMM
REG_WE
REG_SRC
IMM_TYPE
ALU_SRC_PC
ALU_SRC
ALU_CTRL
UNSIGNED
WRD
HWRD
MEM_WRITE
PC_ADD_IMM
CONTROL SIGNALS
func7
func3
opcode
LT
LTU
EQ
Instruction Type
Ctrl Color Code
R-TYPE
I-TYPE
S-TYPE
SB-TYPE
U-TYPE
UJ-TYPE
GENERAL
CLK
RST
Note: Only lower 16 bits used for IMEM. (last 2 bits dropped when shifting right)
PC
NEXT
J_REG
PC_ADD_IMM
IMEM
MEM CTRL
WRITE
I_ADDR
O_ADDR
MEMORY
BE
WE
ADDR
DATA
Programming Ports
IMEM_ADDR
IMEM_WE
IMEM_DATA
0x4
REGFILE
RS1
RS2
RD
I_DATA
O_DATA1
O_DATA2
REG_SRC
ALU
ALU_SRC
IMM CTRL
IMM_TYPE
DMEM
PC+4
Note: LSB is default
SUB WORD WRITE
SUB WORD READ
sh
sw
UNSIGNED
ih
iw
ADDR